

केन्द्रीय विद्यालय संगठन, कोलकाता संभाग
KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION
प्री-बोर्ड परीक्षा / PRE – BOARD EXAMINATION – 2025-26

कक्षा /CLASS - X
विषय /SUBJECT - SCIENCE

अधिकतम अंक /MAXIMUM MARKS - 80
समय /TIME – 3 घंटे / 3 HOURS

ANSWERS AND MARKING SCHEME

SECTION – A

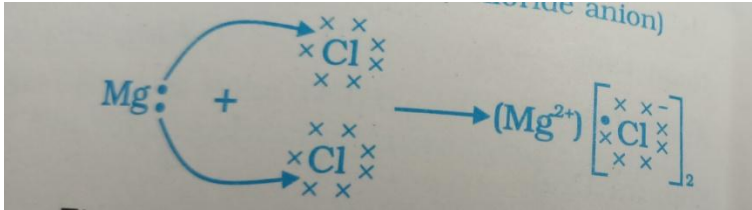
Q.NO	ANSWERS	MARKS
1	c) Nostrils – Pharynx – Larynx – Trachea – Alveoli	1
2	c) Cuscuta	1
3	d) (ii) and (iii)	1
4	b) Neuromuscular junction	1
5	c) 100% ; 75%	1
6	b) (ii) , (iii) and (iv)	1
7	b) Decomposers	1
8	a) Both A and R are true, and R is the correct explanation of A	1
9	d) A is false but R is true	1

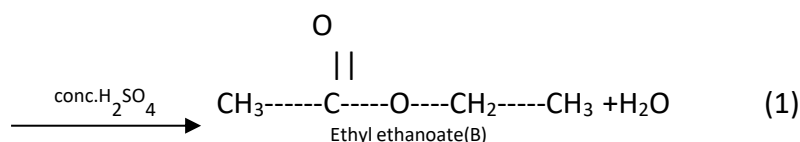
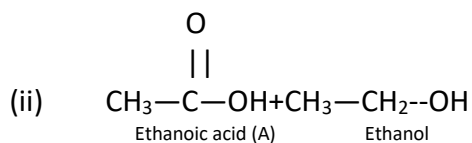
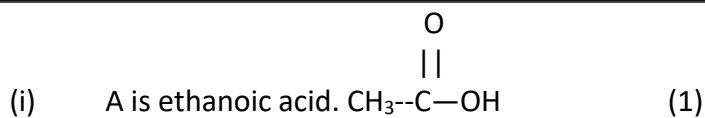
10	i) Gaseous waste in plants is removed through stomata in leaves. (1) ii) Many plants store waste materials in the vacuoles of mesophyll cells and epidermal cells of leaves. When old leaves fall these waste materials are excreted along with the leaves (1)	2
11	A) Left atrium, Pulmonary vein (1+1) OR i) If xylem tissue in a plant is removed the transport of water in the plant will stop. (1) ii) If we are injured, the platelets will clot the blood at the point of injury. (1)	2
12	For primary consumers 10% of 10000 J = 1000 J (1) For secondary consumers 10% of 1000 J = 100 J (1)	2
13	Reflex action is defined as an unconscious, automatic and involuntary response of effectors, i.e., muscles and glands, to a stimulus, which is monitored through the spinal cord. (1) Flow chart :- (2) Unwanted particles in the nose ----- nose ----- sensory neurons --- spinal cord ----- (stimulus) (receptor organ) Motor neuron ----- muscles of face ----- sneeze (effector) (response)	3
14	i) The colour of all the flowers in F_1 generation will be blue. (1) ii) The percentage of white flower plants in F_2 generation will be 25. (1) iii) The ratio of genotype BB and Bb in F_2 progeny will be 1:2. (1)	3
15	i) (a) Renal artery (1/2) (b) Glomerulus ($\frac{1}{2}$) ii) Urinary bladder stores urine. This is a muscular organ under nervous control. ($\frac{1}{2}$ + 1/2) iii) Two major steps involved in the formation of urine are Filtration – In filtration, nitrogenous waste like urea and uric acid are filtered by the glomerulus of kidney (1) Reabsorption – Various useful substances like glucose, salts, amino acids and major amounts of water are reabsorbed by the renal tubules from the initial filtrate. (1) OR iii) In the tubular part of nephron, selective reabsorption takes place. The amount of water reabsorbed depends on following two factors ■ How much excess water is there in the body. (1) ■ How much of dissolved waste is to be excreted (1)	4

16	i) Sepals/calyx and petals/corolla ($1/2 + \frac{1}{2}$) ii) Self pollination: transfer of pollen grain from anther to stigma in the same flower or another flower of the same plant. (1) Cross pollination: transfer of pollen grain from anther to the stigma of one flower to another of two different plants. (1) Significance: (any two) ($1/2 + \frac{1}{2}$) 1 Necessary for seed germination. 2 Stimulates development of fruits. 3 Cross pollination brings about genetic variation 4 Leads to fertilisation iii) Ovule- seed, Ovary- fruit. ($1/2 + \frac{1}{2}$) OR i) Testes are located outside the abdominal cavity in scrotum because sperm formation requires a lower temperature than the normal body temperature. ($1/2+1/2$) Testes function: (a) Production of sperms ($1/2$) (b) Testosterone secretion ($1/2$) ii) (a) The uterine wall thickens and is richly supplied with blood. A special tissue called placenta develops which connects the embryo to the uterine wall and provides nutrients and oxygen to it from the mother's blood, and also removes waste generated by the developing embryo (1) (b)The thick and spongy lining of the uterus slowly breaks and comes out through the vagina as blood and mucous, this process is called menstruation (1) iii) During the surgical method, the vas deferens in the male is blocked thus preventing sperm transfer (Vasectomy). ($1/2$) Similarly, in females, the fallopian tube is blocked, so the egg does not reach the uterus (Tubectomy). Hence, fertilisation will not take place thereby preventing pregnancy ($1/2$)	5
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SECTION – B

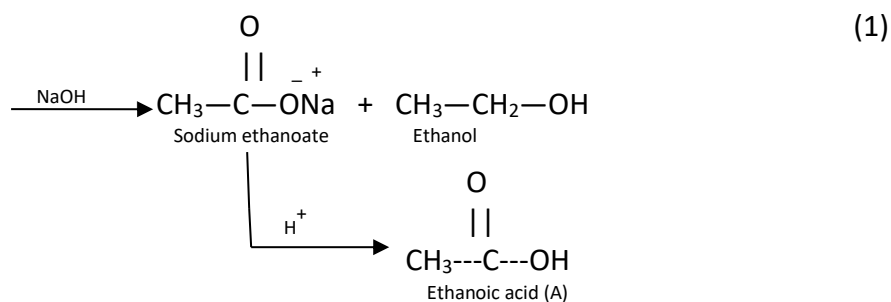
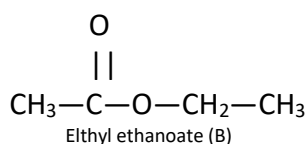
17	c) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$	1
18	b)(ii) and (iii)	1
19	(a) Yes, because iron is more reactive than copper	1
20	B – Red , yellow	1
21	(d)Both (b) and (c)	1
22	(b)Yellow precipitate was formed	1
23	(c)White, BaSO_4	1

24	a) Both A and R are true, and R is the correct explanation of A	1
25	“Displacement reactions also plays a key role in extracting metals in the middle of the reactivity series”. The following two examples Justify this statement a) When iron (III) oxide is reduced by aluminium , the following displacement reaction takes place where aluminium ,being more reactive , displaces iron from its oxide : $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \longrightarrow 2\text{Fe}(\text{l}) + \text{Al}_2\text{O}_3(\text{s}) + \text{heat} \quad (1)$ b) When manganese dioxide is heated with aluminium powder , it is reduced to metallic manganese as shown : $3\text{MnO}_2(\text{s}) + 4\text{Al}(\text{s}) \longrightarrow 3\text{Mn}(\text{l}) + 2\text{Al}_2\text{O}_3(\text{s}) + \text{heat} \quad (1)$	2
26	(i) (a) Electrolytic reduction is used in case of Al, Na. (1) (b) Reduction with carbon is done in case of Zn, Fe, Sn. (1)  (ii) OR (i) It is concentrated by froth- flotation process. (1) (ii) Roasting: The concentrated sulphide ore is heated strongly in the presence of oxygen to convert into its oxide. $2\text{ZnO}(\text{s}) + 3\text{O}_2(\text{g}) \longrightarrow 2\text{ZnO}(\text{s}) + 2\text{SO}_2(\text{g}) \quad (1)$ Reduction: This oxide of metal is reduced with suitable reduction agent to get free metal. $\text{ZnO}(\text{s}) + \text{C}(\text{s}) \xrightarrow{\text{heat}} \text{Zn}(\text{s}) + \text{CO}(\text{g}) \quad (1)$	3
27	(a) The reaction will be much faster, hydrogen gas will be produced more rapidly due to the increased surface area of the zinc dust (1) (b) No reaction will occur, and no hydrogen gas will be produced because copper is less reactive than hydrogen and cannot displace it from dilute sulphuric acid (1) (c) A reaction will occur, but no hydrogen gas will be evolved because nitric acid is a strong oxidizing agent that oxidizes the hydrogen produced to water (1)	3
28	a) The type of reaction is a neutralisation reaction. The products formed are sodium chloride , carbon dioxide and water. (1 + 1) b) The lab assistant sprinkled baking soda over the hydrochloric acid spill to neutralise the strong acid . (1) OR b) safety measures that the teacher might have explained to the students while handling acids : -- Always wear personal protective equipment like safety goggles , gloves and a lab court to protect your skin , eyes and clothing. (1/2) -- Never add water to a concentrated acid instead add acid slowly to water to avoid splashing and dangerous heat build up (1/2) c) The brisk effervescence is caused by the evolution of carbon dioxide gas during the reaction between hydrochloric acid and baking soda. (1)	4

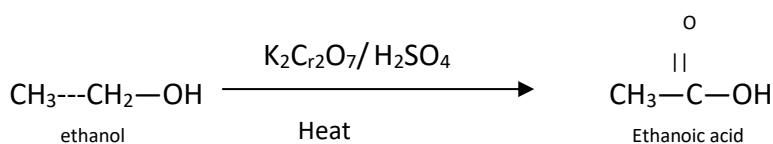


Concentrated H_2SO_4 acts both as a catalyst to speed up the reaction, and a dehydrating agent to remove water.

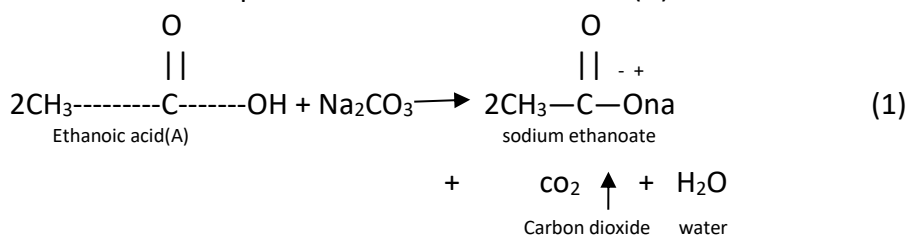
- (iii) Ethyl ethanoate (B) on treatment with sodium hydroxide (an alkali) gives sodium ethanoate and ethanol. This reaction is known as saponification reaction. Sodium ethanoate on further treatment with acid gives back ethanoic acid (A).



- (iv) Ethanoic acid can be obtained from ethanol by oxidation using acidified potassium dichromate as an oxidising agent. (1)



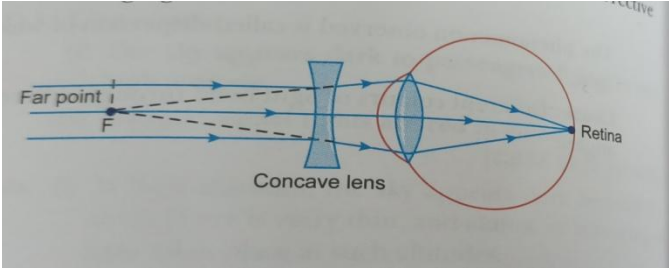
- (v) Carbon dioxide is produced when ethanoic acid (A) reacts with washing soda.



	OR (i) Ethanol, C₂H₅OH (1) (ii) (a) $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \longrightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2$ (1) Sodium ethoxide (b) $\text{C}_2\text{H}_5\text{OH} \xrightarrow[443\text{K}]{\text{Conc. H}_2\text{SO}_4} \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$ (1) Ethene (c) $\text{C}_2\text{H}_5\text{OH} + \text{CH}_3\text{COOH} \xrightarrow{\text{Acid catalyst}} \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$ (1) Ethyl ethanoate (d) $\text{C}_2\text{H}_5\text{OH} \xrightarrow[\text{Heat}]{\text{Acidified K}_2\text{Cr}_2\text{O}_7} \text{CH}_3\text{COOH}$ (1) Ethanoic acid	
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SECTION – C

30	(c) Virtual, upright and diminished in size	1
31	(b) Refraction of light by different layers of varying refractive indices	1
32	(a) Both A and R are true, and R is the correct explanation of A	1
33	(i) Here, u = -20 cm, h = 5 cm, f = -18 cm (1/2) (a) Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\rightarrow \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{-18} + \frac{1}{-20} = \frac{-20-18}{360} = \frac{-38}{360}$ Therefore, v = -360/38 = -9.47 cm Here -ve sign shows that image is formed at the same side of the object. (b) Magnification, m = v/u = -9.47/-20 = 0.47 (1/2) Here +ve sign shows that the image is virtual and erect (ii) The virtual image formed by a concave lens is always diminished while virtual image formed by a convex lens is always magnified (1/2 + 1/2)	2
34	(i) The 10 Ω and 50 Ω resistors are connected in series. Their combined resistance (R_s) is the sum of their individual resistances R_s = 10Ω + 50Ω = 60Ω $\frac{1}{R_{\text{Total}}} = \frac{1}{R_s} + \frac{1}{R_3}$ $\frac{1}{R_{\text{Total}}} = \frac{1}{60} + \frac{1}{30} = \frac{3}{60}$ R_{Total} = 60/3 = 20Ω (1) (ii) Current flowing through the ammeter = I_{Total} = V/R_{Total} = 6/20 = 0.3 A (1)	2

	OR i) Effective resistance of the circuit (1) $R_s = R_3 + R_4 = 4 + 6 = 10\Omega$ $1/R_p = 1/R_s + 1/R_2 = 1/10 + 1/10 = 2/10 = 1/5$ $R_p = 5\Omega$ Total resistance of the circuit $= R_1 + R_p + R_s = 5 + 5 + 10 = 20\Omega$ ii) Current drawn from the battery (1) $V = 20V$, $R = 20\Omega$ $I = V/R = 20/20 = 1A$	
35	The person is suffering from myopia as person suffering from myopia can see nearby objects clearly but cannot see distant objects clearly (i.e. far point of the eye nearer than infinity) (1/2) Causes of myopia: (i) Focal length of the eye lens becomes small (1/2) (ii) The eye ball becomes elongated, so the image of the distant objects is formed in front of the retina (1/2) The corrective lens used for this defect is concave lens. (1/2) Ray diagram showing the function of the corrective diverging lens (1) 	3
36	(i) $P = V \times I$ (1) $I = 1100/220 = 5A$ (ii) $P \times t$ (1) Given $P = 1100W$ Daily usage time = 5 hours Number of days = 6 days $P = 1100/1000 = 1.1KW$ $t = 5 \times 6 = 30$ hours $E = 1.1KW \times 30$ Hours $E = 33KWh$ (iii) energy consumed = 33 unit (1 unit = 1KWh) (1) Rate per unit = Rs 5 Total cost = $33 \times 5 = Rs 165$	3
37	(i) Right-hand Thumb Rule (1) (ii) Fleming's Left-hand Rule (1) (iii) Fleming's Right-hand Rule (1)	3

38	(i) Two parameters change. They are wavelength and velocity. (1) (ii) Yes, it bends towards normal. (1) OR Snell's law states that the ratio of sine of the angle of incidence to the sine of the angle of refraction is a constant for a given colour of light and a given pair of media $\sin i / \sin r = \text{constant}$ (iii) Refractive index of a medium is given by, $n = \text{Speed of light in vacuum} / \text{speed of light in medium} = c/v$ $v = c/n = 3 \times 10^8 \text{ m/s} / 1.5 = 2 \times 10^8 \text{ m/s}$ (2)	4
39	a) Disadvantages : i) All appliances do not get same potential in series arrangement. (1) ii) All appliances cannot be individually operated. (1) b) Let $R_1 = 4\Omega$, $R_2 = 6\Omega$, $R_3 = 15\Omega$, $R_4 = R_5 = 3\Omega$ (3) Equivalent resistance in series $R_{s1} = R_1 + R_2 = 6 + 4 = 10\Omega$ Equivalent resistance in series $R_{s2} = R_4 + R_5 = 3 + 3 = 6\Omega$ Now, R_{s1}, R_3 and R_{s2} are in parallel. Therefore, total resistance of the circuit is $1/R = 1/R_{s1} + 1/R_{s3} + 1/R_{s2}$ $= 1/10 + 1/15 + 1/6$ $= 10/30 = 1/3$ Therefore $R = 3\Omega$ So effective resistance between A and B in the given circuit = 3Ω OR a) Total resistance of series combination is (2) $R = R_1 + R_2 = 1 + 2 = 3\Omega$ Current through this combination, by using Ohm's law $I = V/R = 6/3 = 2\text{A}$ Therefore power consumed in 2Ω resistor is $P = I^2 R = 2^2 \times 2 = 8\text{W}$ b) i) $I = 6/(1+2) = 6/3 = 2\text{A}$ (1/2) Since current flowing is same in both resistors, power used in 2Ω resistor $P_1 = I^2 R = 2^2 \times 2 = 8\text{W}$ ii) Since both 12Ω and 2Ω are in parallel to the 4V source, Power used in 2Ω resistor (1/2) $P_2 = V^2/R = 4^2/2 = 16/2 = 8\text{W}$ Comparison between the power used in both cases is $P_1/P_2 = 8\text{W}/8\text{W} = 1$	5